

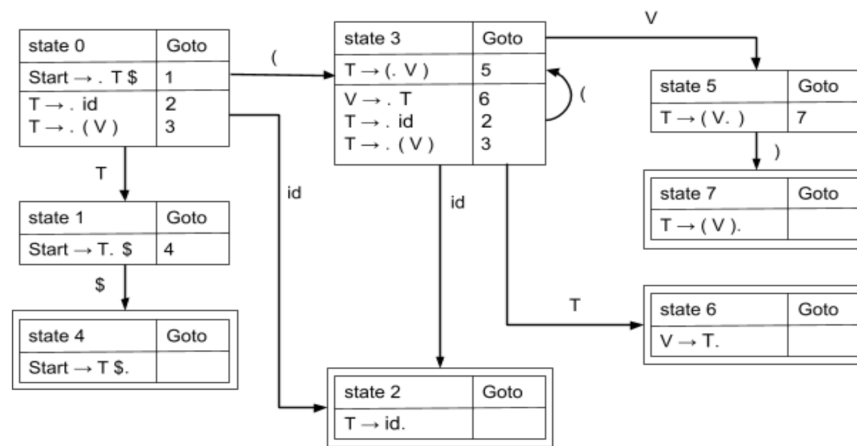
Full Name : _____ Student's ID : _____

Bottom-Up Parsing

Given the following grammar:

- $Start \rightarrow T \$$
- $T \rightarrow id$
- $T \rightarrow (V)$
- $V \rightarrow T$

1. Please show the DFA of LR(0) parsing table. [30pts.]



2. Please show the LALR(1) transition table. [70pts.]

Hint: You may not use up all the rows.

a. Propagation Graph Table [30pts.]

State	LR(0) Item	Goto State	Prop Edges Placed by Step		Initialize ItemFollow	
			E-goto State	Empty Derived	First(γ)	I-goto State
0	1 $Start \rightarrow . T\$$	1	4		\$	2,3
0	2 $T \rightarrow . id$	2	5			
0	3 $T \rightarrow . (V)$	3	6			
1	4 $Start \rightarrow T . \$$	4	10			
2	5 $T \rightarrow id .$					
3	6 $T \rightarrow (. V)$	5	11		)	7
3	7 $V \rightarrow . T$	6	12	8,9		
3	8 $T \rightarrow . id$	2	5			
3	9 $T \rightarrow . (V)$	3	6			
4	10 $Start \rightarrow T\$.$					
5	11 $T \rightarrow (V .)$	7	13			
6	12 $V \rightarrow T .$					
7	13 $T \rightarrow (V) .$					

b. Propagation Pass Table [20pts.]

Item	Prop To	Initial	Pass1	Pass2	Pass3	Pass4	Pass5	Pass6
1 $Start \rightarrow . T \$$	4	\$	\$	\$	\$	\$	\$	
2 $T \rightarrow . id$	5	\$	\$	\$	\$	\$	\$	
3 $T \rightarrow . (V)$	6	\$	\$	\$	\$	\$	\$	
4 $Start \rightarrow T . \$$	10		\$	\$	\$	\$	\$	
5 $T \rightarrow id .$			\$	\$(,)	\$(,)	\$(,)	\$(,)	
6 $T \rightarrow (. V)$	11		\$	\$(,)	\$(,)	\$(,)	\$(,)	
7 $V \rightarrow . T$	8,9,12	)	)	)	)	)	)	
8 $T \rightarrow . id$	5		)	)	)	)	)	
9 $T \rightarrow . (V)$	6		)	)	)	)	)	
10 $Start \rightarrow T \$.$				\$	\$	\$	\$	
11 $T \rightarrow (V .)$	13			\$	\$(,)	\$(,)	\$(,)	
12 $V \rightarrow T .$			)	)	)	)	)	
13 $T \rightarrow (V) .$					\$	\$(,)	\$(,)	

c. LALR(1) Parsing Table [20pts.]

state	id	(	\$	)	Start	T	V
0	S2	S3			acc	S1	
1			S4				
2			Rb	Rb			
3	S2	S3				S6	S5
4			Ra				
5				S7			
6				Rd			
7			Rc	Rc			